

Professional Experience

AI/ML Engineer, Lockheed Martin Space
Littleton, Colorado (2021 – Present)

- Generated over \$1M in research funding through technical proposals for novel research projects
- Principal Investigator in creating AI algorithms for multimodal geospatial recognition that played a pivotal role in securing a multimillion-dollar government contract
- Initiated the integration of Unity ML Agents with Graph Neural Network (GNN) for a multi-agent reinforcement learning (RL) project aimed at enhancing autonomous swarm systems
- Technical Lead that helped secure Lockheed's first commercial environmental monitoring contract for The Occidental Petroleum Company
- Developed a novel transformer architecture to handle the sparse event data from neuromorphic sensors for classification and segmentation tasks
- Lead a research team using AI for novel polymer generation to improve material science discovery processes for high temperature space materials
- Integration Lead for the Data Sensor Fusion project integrating frontend and backend systems to enable real-time data processing using React, MySQL, and Flask

Signals & Image Processing Engineer Asc., Lockheed Martin Space
Littleton, Colorado (2020 – 2021)

- Implemented deep learning inference pipelines using ONNX and TensorRT for overhead computer vision, enhancing inference speeds by 10x critical customer needs
- Object level change detection (OLCD) project lead where I used React, vector databases, and advanced AI algorithms for activity recognition across modalities
- Researched combining quantum computing to use with deep learning for fingerprinting high value object detection

Software Development Intern, Lockheed Martin Space
Littleton, Colorado (2018 – 2019)

- Supported the Synthetic Aperture Radar Automatic Target Recognition (SAR ATR) team with integrations between F35 software and neural network software
- Built a Generative Adversarial Network (GAN) that creates SAR images to enlarge small datasets
- Developed a satellite vehicle recognition software using Convolutional Neural Networks (CNNs) with TensorFlow
- Authored a white paper that proposed using large language models (LLMs) to forecast emerging technology trends in academia to make the internal company research selection process data driven

Lead Software Architect, Rocketry@VT
Blacksburg, Virginia (2016 – 2019)

- Designed active drag system software so that the rocket would hit an exact target altitude that helped secure a top 10 worldwide finish at the Spaceport America Cup
- Developed a real time aerial target tracking system in Pytorch to identify objects during flight

Research Publications

- Object-level change detection for autonomous sensemaking*: SPIE Defense + Commercial Sensing 2022
- Optimizing deep learning classifier performance for low to medium resolution overhead imagery*: SPIE Defense + Commercial Sensing 2021
- A comparative study of motor imagery based BCI classifiers on EEG and iEEG Data*: GlobalSIP 2019

Awards

- 2023 Innovation Leader*: Recognized for the creation of multiple trade secrets and leading key projects
- LM Space Project Innovation Award: OLCD 2022*
- 'Politigauge': an NLP powered political sentiment analysis web extension that secured first place at the Virginia Tech Undergraduate Computer Science Research Symposium 2019*

Education

Georgia Tech, M.S. Machine Learning
Virginia Tech, Magna Cum Laude, B.S. Computer Science

Expected 2025
December 2019

Technical Skills

Python	AWS	C#	AI/ML	CI/CD	Pytorch	SQL	C++	Blender
MATLAB	JavaScript	TensorFlow	Docker	Git	Linux	MongoDB	Unity	Godot